

RGPV PREVIOUS YEAR QUESTIONS (PYQ) ANALYSIS REPORT

Course: B.Tech. I/II Semester • Subject: Basic Mechanical Engineering (BT-203 GS)

Complete Unit-Wise Examination Analysis & Trend Guide

Comprehensive Analysis of Exam Sessions: June 2022 – December 2025

Target Audience: First-Year B.Tech. Students (Grading System)

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Table of Contents

1. Section 1: Unit-wise Previous Year Questions (PYQs)	3
• Unit 1: Materials & Mechanical Properties	3
• Unit 2: Measurement, Metrology & Machine Tools	3
• Unit 3: Fluid Mechanics & Fluid Machinery	4
• Unit 4: Steam Boilers & Thermodynamics	5
• Unit 5: IC Engines, Cycles & Air Compressors	5
2. Section 2: Unit-wise Important Topics & Priority Matrix	7
3. Section 3: Most Repeated PYQs Ranked by Frequency	10

Section 1: Unit-wise PYQs

This section lists every question from the RGPV Basic Mechanical Engineering (BT-203 GS) examination papers from June 2022 to December 2025. Questions have been transcribed exactly as they appear and sorted into their correct curriculum units.

Unit 1: Materials & Mechanical Properties

1. Define alloy steel. Explain the effect of alloying elements on steel properties. (Dec 2025)
2. Describe important phases and critical points in the Iron-Carbon diagram with the help of sketch. (Dec 2025)
3. Classify engineering materials. What do you mean by alloys? Write some applications of alloys. (June 2025, Nov 2022)
4. Define the following Mechanical properties: (i) Toughness, (ii) Brittleness, (iii) Tensile strength, (iv) Ductility. (June 2025)
5. What is the necessity of Alloying? What are alloy steels, and how do they differ from carbon steels? (Dec 2024)
6. What is the purpose of hardness testing in materials and how it is performed? (Dec 2024)
7. Write in detail classification of cast irons along their applications. (Dec 2024)
8. Explain in detail Classification of engineering material along their applications. (Dec 2024)
9. Define the following Mechanical properties of materials: (i) Strength, (ii) Hardness, (iii) Toughness, (iv) Ductility, (v) Brittleness, (vi) Malleability. (June 2024)
10. Draw Fe-Fe₃C phase diagram and explain the phase transformation reactions in the diagram. (June 2024)
11. Define Hardness? Explain the Brinell Hardness test method in details? (Dec 2023)
12. Define the following mechanical properties: (i) Strength, (ii) Ductility. (Dec 2023)
13. Explain the effect of any 8 (eight) alloying element in alloy steel with details. (June 2023)
14. Enlist important mechanical properties of engineering materials. Define any three with suitable examples. (June 2022)
15. Explain the methods to measure the hardness of materials. (Nov 2022)

Unit 2: Measurement, Metrology & Machine Tools

1. Define lathe. Explain the main parts of a lathe machine with neat sketch. (Dec 2025)
2. List the different types of temperature measuring instruments. Also explain the working of optical pyrometer. (Dec 2025)
3. Describe tensile testing and explain stress-strain diagram of ductile materials. (Dec 2025)
4. Define modulus of elasticity. Explain Young's modulus, bulk modulus and shear modulus. (Dec 2025)
5. Explain the construction and working principle of a dial gauge with its applications. (Dec 2025)
6. Describe the construction and use of sine bar. How will you measure taper of a job with the help of a sine bar. (June 2025)
7. Describe the procedure of a tensile test. Explain the stress-strain curve for a ductile material. (June 2025, June 2023)
8. With a neat diagram explain the main parts of a drilling machine? Explain any two operations performed on drilling machine. (June 2025, Dec 2024, June 2022)
9. Explain working of a thermocouple and radiation pyrometer with neat sketches. (June 2025)
10. Write short notes on: (i) Vernier caliper, (ii) Micrometer, (iii) Dial gauge. (Dec 2024)
11. Describe the typical shape of the stress-strain curve for a ductile material. Explain the significance of the yield point and ultimate tensile strength on the stress-strain curve of a ductile material. (June 2024)
12. State and explain various types of errors in measurements. (June 2024)
13. List out various operations performed by Lathe and explain some of them in detail. (June 2024)
14. List out various Temperature measuring instruments and explain any two with neat diagram. (June 2024)
15. Write short notes on Slip gauge, Sine-bar and Combination set. (June 2024)
16. A mild steel specimen with an original diameter of 10 mm and a gauge length of 50 mm was found to have an ultimate load of 60 kN and breaking load of 40 kN. The gauge length at rupture was 55 mm and diameter at rupture cross-section was 8 mm. Determine: (i) The ultimate stress, (ii) Breaking stress, (iii) True breaking stress, (iv) Percentage elongation. (Dec 2023)
17. With a neat sketch, explain the following lathe operations: (i) Facing, (ii) Knurling. (Dec 2023)
18. Discuss the application vernier caliper and its method of measurement. (Dec 2023)
19. Explain the method of using Sine Bar with neat sketch. (Dec 2023)
20. Explain feed mechanism used in a drilling machine. (June 2023)

21. Explain the method of using Slip Gauge with the neat sketch. (June 2023)
22. Explain the experimental set up of tensile testing of steel. Also compare stress-strain diagrams of ductile and brittle materials. (June 2022)
23. Find the young's modulus of a brass rod of diameter 25 mm and of length 250 mm which is subjected to a tensile load of 50 kN when the extension of the rod is equal to 0.3 mm. (June 2022)
24. What are the various methods used for temperature measurement? Explain any one of them. (June 2022)
25. Define welding. Classify welding processes. Explain arc welding process. (June 2022)
26. With a neat diagram explain the main parts of a lathe machine. Explain any two operations performed on a lathe machine. (Nov 2022)
27. Explain the measurement process using vernier calliper with neat sketch. (Nov 2022)
28. Short notes on: Micrometer (June 2025, Dec 2023, June 2022), Dial gauge (June 2022, Nov 2022), Sine bar (Nov 2022), Hooks law and modulus of elasticity (Dec 2023), Casting and Patterns (June 2023), Impact testing of materials (June 2023), Casting defects (June 2022), Welding defects (Nov 2022).

Unit 3: Fluid Mechanics & Fluid Machinery

1. Derive Bernoulli's equation for incompressible fluids. Also discuss its physical significance. (Dec 2025, Dec 2024, June 2022)
2. Explain the construction and working principle of a reciprocating pump. (Dec 2025, Dec 2023, June 2022)
3. Explain the following: (i) Homogeneous and heterogeneous fluids, (ii) Newton's law of viscosity. (Dec 2025)
4. The water is flowing through a pipe having diameters 25cm and 10cm at section 1 and 2 respectively. The rate of flow of water through the pipe is 30 lit/sec. Section 1 is 10m above the ground and section 2 is 2m above the ground. If pressure at section 1 is 40 N/cm², find the pressure at section 2. (June 2025)
5. Describe the construction and working principle of the pelton wheel? (June 2025, June 2023, Nov 2022)
6. What is the basic principle of turbine? Classify various types of turbines according to type of energy, direction of flow head and specific speed. (June 2025)
7. Explain how hydraulic turbines convert fluid energy into mechanical energy. Discuss the types of hydraulic turbines and their applications. (Dec 2024)
8. Define and distinguish between: (i) Uniform flow and non-uniform flow, (ii) Laminar and turbulent flow, (iii) Newtonian and Non-Newtonian flow. (Dec 2024)

9. State Newton's law of viscosity and explain its significance. Describe how viscosity is related to shear stress and shear rate in Newtonian fluids. (Dec 2024)
10. Describe the working principle of hydraulic pumps. Give classification of pumps with their applications. (June 2024)
11. What is Pascal's law, Explain how Pascal's law is applied in hydraulic systems? (June 2024, June 2022)
12. The water is flowing through a pipe having diameters 30 cm and 20 cm at two sections 1 and 2, respectively. The rate of flow through the pipe is 80 liters/sec. Section 1 is 8 m above the datum and section 2 is 5 m above the datum. If the pressure at section 1 is 4 bar, find the intensity of pressure at section 2. (Dec 2023)
13. What is priming? Why is it required in a centrifugal pump? (Dec 2023)
14. Explain the process of Pitot tube for velocity measurement of a Fluid with neat sketch. (June 2023)
15. The dynamic viscosity of lubricating oil used between shaft and sleeve is 10 poise. The diameter of the shaft, which rotates at 200 rpm, is 0.5 m. The sleeve length is 100 mm. Calculate the power lost if the thickness of oil film is 2 mm? (June 2023)
16. What is a draft tube? Why is it used in a reaction turbine? Describe with a neat sketch? (June 2023)
17. A plate, 0.025mm distant from a fixed plate, moves at 60 cm/s and requires a force of 2 N/m² to maintain this speed. Determine the fluid viscosity between the plates. (June 2022)
18. A flat plate of area 1.5×10^6 mm² is pulled with a speed of 0.4 m/s relative to another plate located at a distance of 0.15 mm from it. Find the force and power required to maintain this speed, if the fluid separating them is having viscosity as 1 poise. (Nov 2022)
19. Define Viscosity. What is the importance of viscosity while selecting fluids in various engineering applications? (Nov 2022)
20. Short notes on: Newton's law of viscosity (June 2025, Dec 2023, Dec 2025), Different types of fluids (June 2022).

Unit 4: Steam Boilers & Thermodynamics

1. Define boiler. Explain the classification of boilers also explain any one of them. (Dec 2025)
2. Define draught. Explain natural draught and its working. (Dec 2025)
3. Define thermodynamic system and explain open, close and isolated systems. (Dec 2024, June 2022)
4. Differentiate between boiler mountings and accessories. Discuss the role of boiler accessories in enhancing efficiency and safety in boiler systems. (Dec 2024)

5. What does the First Law of Thermodynamics state, and how does it relate to the conservation of energy? (Dec 2024)
6. What are the main types of boilers, and how are they classified based on construction and operation? (June 2024)
7. Explain the concept of draught in boiler operation. Discuss the factors influencing the choice between natural and artificial draught systems in boiler installations. (June 2024)
8. In a boiler test the following quantities were obtained; mean temperature of the feed 20°C ; mean boiler pressure 10 kgf/cm^2 , mean steam dryness 0.9, coal burnt/hr 250kg, water supplied to boiler per hour 2260 kg. The mass of water in the boiler at the end of the test was less than that at the commencement by 240 kg. Calculate: (i) The actual evaporation per kg of coal and (ii) The equivalent evaporation. [Take $h_f=181.3 \text{ kcal/kg}$; $h_{fg}=482 \text{ kcal/kg}$] (June 2022)
9. During a trial operation of a boiler 1630 kg of coal were consumed in 24 hrs and 13000 kg of water evaporated. The mean steam pressure was 7 kgf/cm^2 and condition dry saturated. The fuel contained 3 percent moisture and 4 percent ash and had a calorific value of 7200 kcal/kg. The feed water temperature was 35°C . Determine: (i) The boiler efficiency, (ii) The equivalent evaporation per kg of dry coal, (iii) The equivalent evaporation per kg of combustibles. (Nov 2022)
10. Discuss briefly the differences between externally fired and internally fired steam generators. Also discuss the application of each type of steam generators. (Nov 2022)
11. Discuss, how the steam generators are classified. Give example of each classification. (June 2022)
12. Explain the construction and working of Babcock and Wilcox boiler with a neat sketch. (Dec 2023)
13. Discuss the concept of thermal equilibrium and state zeroth law of thermodynamics? (Dec 2023)
14. Explain the working of Cochran boiler and fusible plug with neat sketches. (June 2023)
15. Derive an expression for the first law of thermodynamics applied to a closed system. Define the internal energy of a system. (June 2023)
16. State the Zeroth Law of thermodynamics. How is mercury in thermometer able to find the temperature of a body using the Zeroth law of thermodynamics? (Nov 2022)
17. Short notes on: Zeroth law of thermodynamics (June 2025), Natural and artificial draught (Dec 2023), Thermodynamic system, properties and state (June 2023), Thermodynamic properties (Nov 2022).

Unit 5: IC Engines, Cycles & Air Compressors

1. Explain the Diesel cycle with P-V and T-S diagram. Also Derive the efficiency of the Diesel cycle. (Dec 2025, Dec 2024)
2. Explain the working of a four-stroke petrol engine with neat sketch. (Dec 2025, Dec 2023)

3. Define compressor. Explain the working principle of a reciprocating air compressor. (Dec 2025, June 2024)
4. Define the following Statements of second law of thermodynamics: (i) Clausius statement, (ii) Kelvin-Planck statement. (Dec 2025, Dec 2023)
5. What is constant volume process? Derive expressions for the work done, change in internal energy heat transfer and change in enthalpy. (June 2025)
6. Define internal energy and prove that energy is a property of a system? (June 2025)
7. State the first law of thermodynamics for a closed system undergoing a cyclic process and non-cyclic process. (June 2025)
8. Explain the structure and working of two stroke diesel engine with a neat sketch? (June 2025)
9. With the help of a neat sketch describe the construction and working of a steam engine? (June 2025)
10. Write about the various processes involved in Dual cycle with P-V and T-S diagrams. (Dec 2024)
11. Explain the working of a 4 stroke diesel engine with neat sketch. (Dec 2024)
12. Explain thermodynamic processes at constant pressure and volume. (June 2024)
13. With a neat sketch, explain the working of Carnot and derive the expression for thermal efficiency. (June 2024, June 2022)
14. Explain the working of a 2 stroke petrol engine with neat sketch. (June 2024)
15. Write short notes on surging, choking and stalling. (Dec 2023)
16. State the Kelvin-Planck and the Clausius statements of the second law of thermodynamics. Explain the equivalence of Kelvin-Planck and Clausius statements. (Dec 2023)
17. Prove that work and heat are the path functions. (June 2023)
18. Explain the structure and working of four-stroke diesel engine with a neat sketch? (June 2023, Nov 2022)
19. Explain the working principle of the centrifugal compressor. (June 2023)
20. Derive an expression for efficiency of Otto cycle with neat sketches of PV and TS diagrams. (Nov 2022)
21. Explain the working of two stroke petrol engine with neat sketch. (June 2022)
22. Short notes on: Centrifugal compressor (June 2025), Working principle of steam Engine (June 2023).

Section 2: Important Topics

This section outlines the recurring high-yield topics determined by examining trends across the 2022–2025 datasets. Use this matrix to guide your high-priority study sessions.

Unit 1: Materials & Mechanical Properties

Topic Category	Priority	Analysis & Exam Expectations
Hardness Testing Methods (Brinell)	★★★ High	Appeared 4 times. Demands complete knowledge of Brinell mechanism, indentation formulas, and general testing objectives.
Alloying Elements & Effects on Steel	★★★ High	Appeared 3 times. Focuses on why alloying is necessary and the distinct structural additions (e.g., Cr, Ni, Mn).
Iron-Carbon Diagram Reactions	★★ Medium	Appeared 2 times. Requires drawing phase boundaries and identifying key phases (Austenite, Ferrite, Cementite, Pearlite).
Mechanical Properties Definitions	★★ Medium	Appeared 3 times. Standard 2-mark definitions covering core terminologies (Ductility, Malleability, Toughness).

Unit 2: Measurement, Metrology & Machine Tools

Topic Category	Priority	Analysis & Exam Expectations
Ductile Stress-Strain Diagram	★★★ High	Appeared 5 times. Core descriptive element; mandatory to plot elastic limit, yield points, necking, and rupture. Includes numerical extensions.
Drilling Machine Parts & Operations	★★★ High	Appeared 3 times. Regularly tests diagram labeling along with specific functional tasks (Boring, Reaming, Tapping).
Lathe Machine Configuration	★★★ High	Appeared 3 times. Focuses on architecture (headstock, tailstock, carriage) and tool-travel processes like facing and knurling.
Linear & Angular Instruments	★★ Medium	Repeated short notes on Micrometer, Vernier, Sine Bar, and Dial Gauge across almost all papers.

Unit 3: Fluid Mechanics & Fluid Machinery

Topic Category	Priority	Analysis & Exam Expectations
Bernoulli's Equation & Applications	★★★ High	Appeared 5 times. High derivation focus for incompressible fluids. Consistently coupled with pipe-flow conservation numericals.
Reciprocating & Centrifugal Pumps	★★★ High	Appeared 5 times. Covers single-acting configurations, operational sketches, and concepts like priming.
Pelton Wheel Turbine Setup	★★★ High	Appeared 3 times. Dominant impulse turbine question testing operational layouts and bucket designs.
Newton's Law of Viscosity	★★ Medium	Appeared 4 times. Demands linear velocity gradient equations matched with plate/fluid shear resistance math.

Unit 4: Steam Boilers & Thermodynamics

Topic Category	Priority	Analysis & Exam Expectations
Boiler Classification & Tests	★★★ High	Appeared 5 times. Demands broad classification systems and numerical mastery over boiler trials (equivalent evaporation, efficiency).
Babcock & Wilcox vs Cochran	★★★ High	Appeared 2 times. The standard fire-tube vs water-tube benchmark systems tested via detailed schemas.
Boiler Mountings vs Accessories	★★ Medium	Appeared 2 times. Differentiates safety regulations from performance-enhancing preheaters/economizers.
Boiler Draught Principles	★★ Medium	Appeared 3 times. Focuses on natural vs mechanical pressure heads.

Unit 5: IC Engines, Cycles & Air Compressors

Topic Category	Priority	Analysis & Exam Expectations
Air Standard Cycles (Diesel & Otto)	★★★ High	Appeared 4 times. Demands direct derivation of cycle efficiencies from P-V and T-S state paths.
4-Stroke Petrol vs Diesel Working	★★★ High	Appeared 4 times. Heavy visual requirement to show valve timings and piston strokes across cycles.
Reciprocating Air Compressors	★★ Medium	Appeared 2 times. Focuses on working cycles and volume compression phases.
Second Law Statements	★★ Medium	Appeared 2 times. Requires Clausius and Kelvin-Planck verbatim definitions and heat-pump equivalency steps.

Section 3: Most Repeated PYQs

This section merges closely related variants and ranks questions by their statistical appearance frequency between 2022 and 2025.

Unit 1: Materials & Mechanical Properties

1. Hardness Measurement and Testing Principles

Sessions: Dec 2024, June 2024, Dec 2023, Nov 2022 | **Frequency: 4 times**

Core Focus: Explaining measurement methods with deep analysis of the Brinell hardness indenter framework.

2. Alloying Necessity and Element Effects on Steel

Sessions: Dec 2025, Dec 2024, June 2023 | **Frequency: 3 times**

Core Focus: Defining differences between carbon and alloy steel, detailing structural transformations driven by additive elements.

3. Iron-Carbon Phase Diagram Transformations

Sessions: Dec 2025, June 2024 | **Frequency: 2 times**

Core Focus: High precision mapping of critical cooling points and eutectoid/peritectic reaction balances.

Unit 2: Measurement, Metrology & Machine Tools

1. Stress-Strain Curve for Ductile Material & Tensile Testing

Sessions: Dec 2025, June 2025, June 2024, June 2023, June 2022 | **Frequency: 5 times**

Core Focus: Continuous structural plotting requirements, analyzing proportional limits, elastic moduli, elongation, and related tensile metrics.

2. Drilling Machine Anatomy and Functional Operations

Sessions: June 2025, Dec 2024, June 2022 | **Frequency: 3 times**

Core Focus: Outlining parts via line sketches and detailing operations like reaming and counterboring.

3. Lathe Machine Parts and Machining Operations

Sessions: Dec 2025, June 2024, Nov 2022 | **Frequency: 3 times**

Core Focus: Sketching construction layout details and illustrating face-turning or knurling procedures.

Unit 3: Fluid Mechanics & Fluid Machinery

1. Bernoulli's Conservation Equation for Incompressible Fluids

Sessions: Dec 2025, Dec 2024, June 2024, Dec 2023, June 2022 | **Frequency: 5 times**

Core Focus: Comprehensive algebraic pressure head derivations, physical significance breakdown, and conservation numerical problem solving.

2. Reciprocating Pump Construction & Operation

Sessions: Dec 2025, June 2024, Dec 2023, June 2022 | **Frequency: 4 times**

Core Focus: Detailing components of single-acting reciprocating displacement models with neat layouts.

3. Pelton Wheel Impulse Turbine Working Principle

Sessions: June 2025, June 2023, Nov 2022 | **Frequency: 3 times**

Core Focus: Outlining nozzle jets, deflector buckets, and general mechanical conversions.

Unit 4: Steam Boilers & Thermodynamics

1. Steam Boiler Classification and Trial Performance Evaluation

Sessions: Dec 2025, June 2024, Nov 2022, June 2022 | **Frequency: 4 times**

Core Focus: Differentiating steam generator varieties and computing efficiency values from testing matrices.

2. Boiler Draught (Natural vs Artificial) Operational Context

Sessions: Dec 2025, June 2024, Dec 2023 | **Frequency: 3 times**

Core Focus: Explaining stack height pressure drops and comparing chimney flow models.

3. Thermodynamic Systems Classifications (Open, Closed, Isolated)

Sessions: Dec 2024, June 2023, June 2022 | **Frequency: 3 times**

Core Focus: Drawing boundaries and mapping mass-energy flow exchanges across boundaries.

Unit 5: IC Engines, Cycles & Air Compressors

1. Diesel Cycle Thermodynamic Efficiency Derivation

Sessions: Dec 2025, Dec 2024 | **Frequency: 2 times**

Core Focus: Mapping constant pressure heat addition, plotting state trajectories on P-V/T-S grids, and deriving thermal efficiency.

2. Four-Stroke Petrol Engine Suction-Compression-Power-Exhaust Cycles

Sessions: Dec 2025, Dec 2023 | **Frequency: 2 times**

Core Focus: Visualizing spark ignition setups and defining valve/piston mechanics over full cycle steps.

3. Reciprocating Air Compressor Principles

Sessions: Dec 2025, June 2024 | **Frequency: 2 times**

Core Focus: Outlining standard displacement models, clarifying intake cycles and compressed air discharge.